Title

Code

Foundational Concepts

The foundation of statistical inference is comparing the results that we nd from actual data that we collect with what we should expect from random processes. But our intuition about what we are supposed to expect from a random process is often far from accurate.

Hide

```
if (!caracas::has_sympy()) {
  caracas::install_sympy()
}
library(caracas)
library(sets)
library(tibble)
library(ggplot2)
library(dplyr)
library(broom)
library(RColorBrewer)
```

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```
def_sym("X", "Y")

cmStats <- modules::use("CmStats.R")$cmStats
```

When every possible outcome of an experiment is equally likely, it is easy to calculate the probability of an outcome. If a coin is equally likely to land heads as tails when ipped, we say the probability of heads is 50% (or 0.5). For a fair die, the probability is 1/6 of rolling a 4, for example. For cases where possible outcomes are not equally likely, the probability of a possible outcome means the fraction of the time that the outcome will occur.

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```
coins <- sample(c("T", "H"), size=100, replace = TRUE)

cmStats$P(table(coins), 100)
```

```
coins
   H    T
0.57 0.43
```

Hide

```
dice <- sample(c(1:6), size=100, replace = TRUE)

cmStats$P(table(dice), 100)
```

```
dice
   1    2    3    4    5    6 
0.26 0.12 0.14 0.10 0.19 0.19 
```

Randomness becomes more subtle the more we think about it. When ipping a coin, in what sense is the outcome a random event? Once we let the coin go, physics determines what the outcome will be. Interestingly, in quantum physics, the basic model of particles is inherently probabilistic.

We humans have bad intuition about probabilities. The following is an example that is surprising to most people. Suppose there are 50 random people in a room. What is the probability that two of them will have the same birthday? The surprising answer is that there is a 97% chance that two of them will have the same birthday.

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```
start_day <- 366 - 50 + 1

prod(mapply(\(v) cmStats$P(v, 366), start_day:366))
```

```
[1] 0.02992693
```

The product of all the fractions is about 0.03. Thus, the probability that no two people have the same birthday is only about 3%. Hence, the chance that two people do have the same birthday is about 97%.

When two fair dice are thrown and summed, not all possible sums are equally likely. Suppose you have a red die and a black die, and each is a fair die. There are six ways of getting the sum 7: red 1, black 6; red 2, black 5; and so on. We can draw a histogram showing how many ways there are to get each of the numbers 2 through 12. The histogram has a peak over the value 7. When three fair dice are thrown and summed, the histogram of possible sums is more peaked than the histogram for two dice.

Hide

```
purrr::map(1:6, \(d) sum(c(d, 7 - d)))
```

```
[[1]]
[1] 7

[[2]]
[1] 7

[[3]]
[1] 7

[[4]]
[1] 7

[[5]]
[1] 7

[[6]]
[1] 7
```

Hide

```
(7 - 1) / prod(c(6, 6))
```

```
[1] 0.1666667
```

Hide

```
cmStats$P(1, 6)
```

```
[1] 0.1666667
```

Hide

```
dice_red <- sample(c(1:6), size=100, replace = TRUE)
dice_black <- sample(c(1:6), size=100, replace = TRUE)

sum(outer(dice_red, dice_black, FUN = "+") == 7)
```

```
[1] 1648
```

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```
counts <- table(purrr::map2_dbl(dice_red, dice_black, \(x, y) x + y))
counts
```

```
 2   3   4   5   6   7   8   9  10  11  12
 3   4   9   8  14  16  23  11   3   6   3
```

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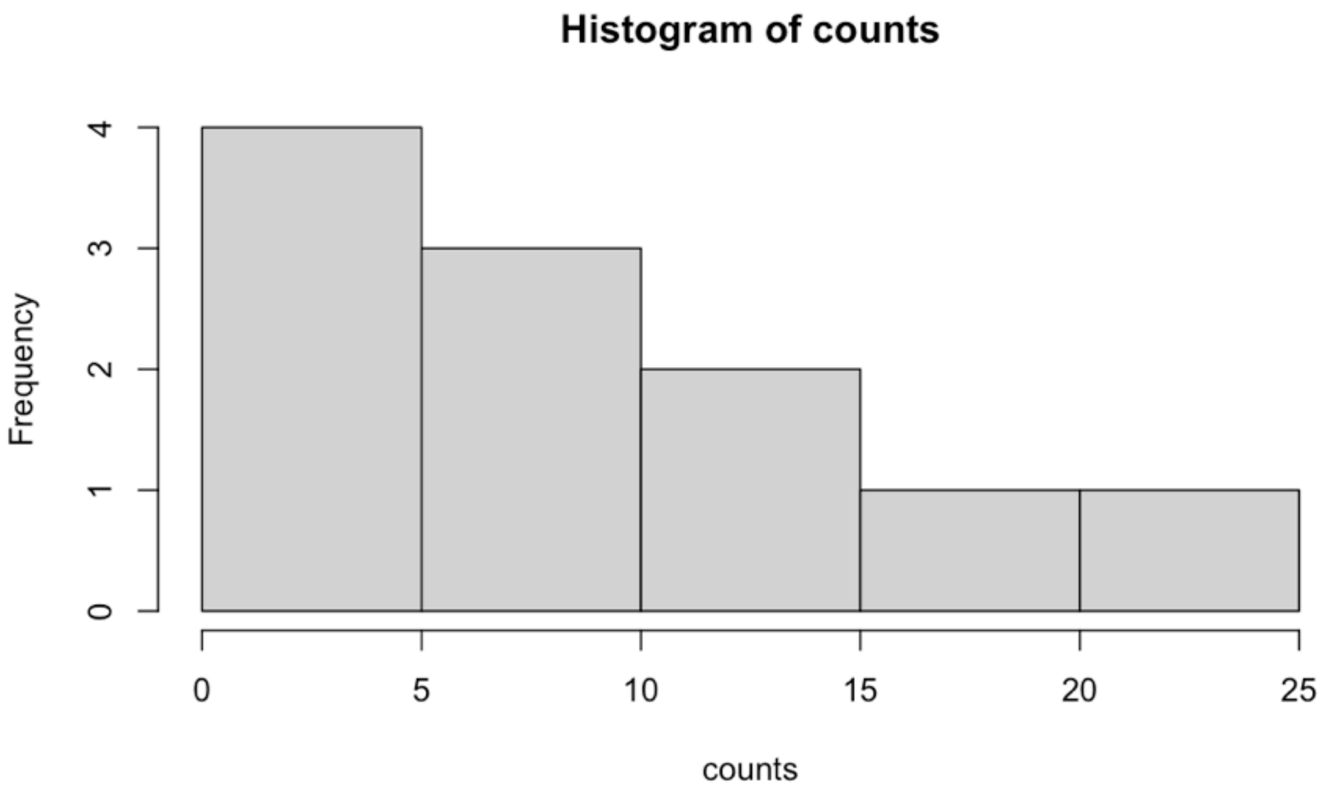
```
hist(counts)
```



Saying that the histogram is more peaked is saying that the standard deviation is smaller. In fact, the standard deviation for the case of 100 dice is 1/10 of the standard deviation for the case of a single die. If we used 1 million dice, we would get an extremely peaked histogram: The standard deviation would be only 1/1000 of the single-die case. The standard deviation decreases proportionately as the square root of the number of dice increases. The fact that the histogram becomes more peaked as the number of thrown dice increases is really an illustration of the central limit theorem.

Hide

```
dice_red <- sample(c(1:6), size=1000, replace = TRUE)
dice_black <- sample(c(1:6), size=1000, replace = TRUE)

sum(outer(dice_red, dice_black, FUN = "+") == 7)
```

```
[1] 166712
```

Hide

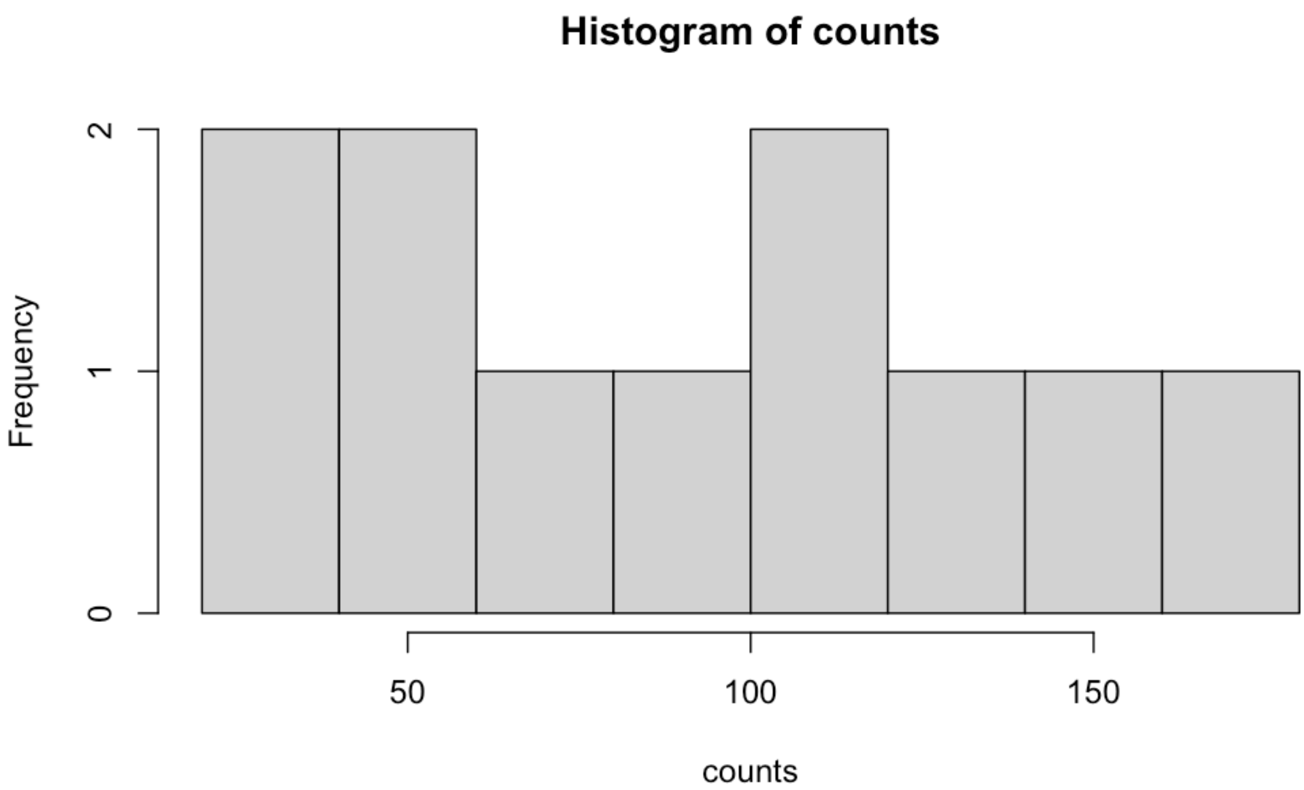
```
counts <- table(purrr::map2_dbl(dice_red, dice_black, \(x, y) x + y))
counts
```

```

  2   3   4   5   6   7   8   9  10  11  12 
 22  59  94 112 138 161 148 101  73  55  37 
```

Hide

```
hist(counts)
```



The central limit theorem implies that for a fairly large sample size n, the distribution of the sample means is close to the normal distribution centered at the same mean as the population mean and with a standard deviation approximately equal to the standard deviation of the original distribution divided by the square root of n. The central limit theorem is one of the workhorses of statistical inference.

These sample means have a distribution that is approximated by a normal distribution.

In the next lectures, we will see how our ability to understand probabilities lies at the heart of the logic of statistical inference.

Concept and Applications

SAMPLE SIZE AND SAMPLING DISTRIBUTIONS

Statisticians are often called on to do consulting, whether for individuals, companies, or nonprofits. This lecture focuses on a consulting example involving a large metropolitan city that is considering building a new hospital to meet the needs of residents. They want to survey the population to better understand the typical emergency room (ER) demand so that they can plan an appropriate number of beds.

SAMPLE MEANS

One thousand residents are sampled and their number of visits to the ER are recorded. The number of times a person visited the ER ranges from 0 to 59 times in a year. The counts for each visit are given in the vector "counts." The vector "visits" repeats each "time" for each corresponding "count."

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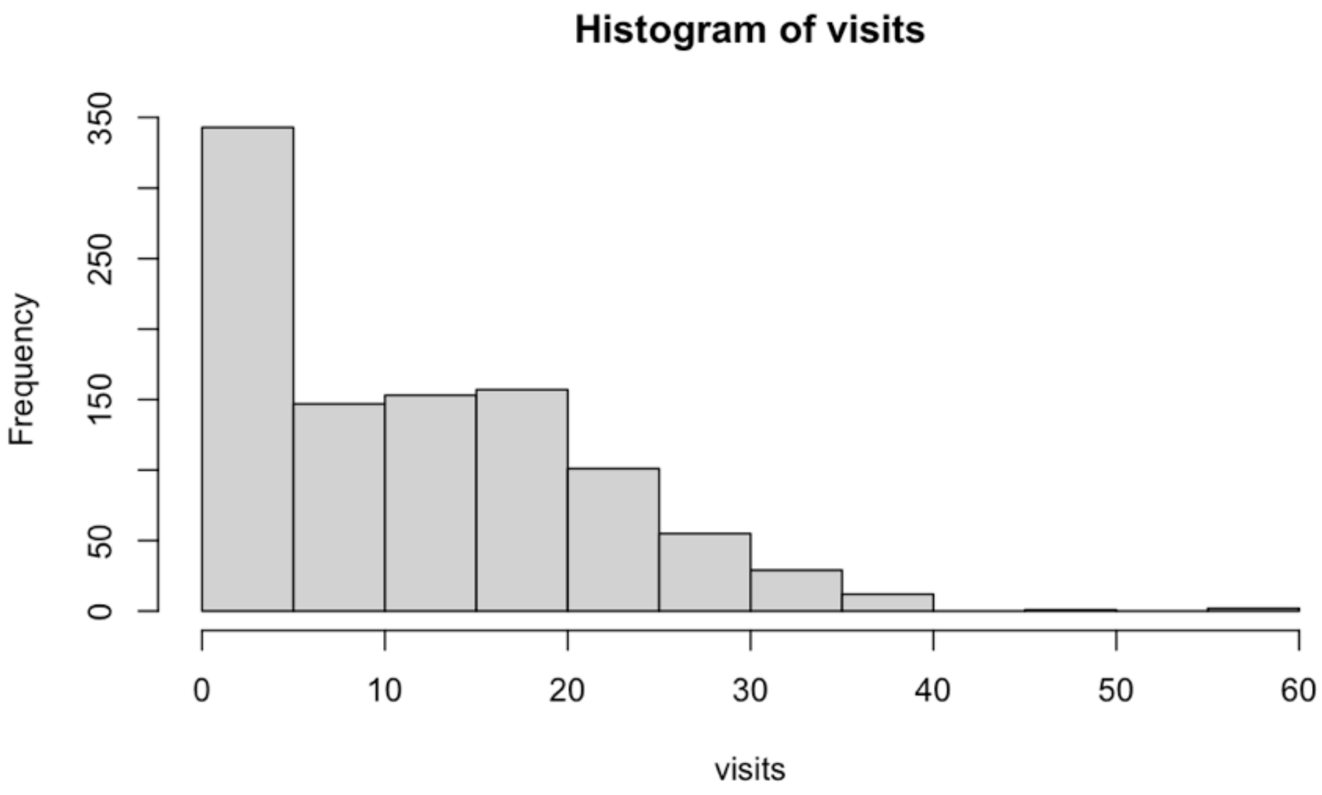
```
times <- (0: 59)

counts <- c(49, 51, 50, 85, 47, 61, 29, 29, 32, 21, 36,
38, 30, 27, 24, 34, 38, 37, 24, 32, 26, 23, 27, 22, 19,
10, 10, 12, 13, 8, 12, 5, 6, 6, 1, 11, 2, 4, 2, 1, 3, 0,
0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1)

visits <- rep.int(times, counts)
```

The following histogram displays information about the distribution of visits for our population of 1000 residents. Notice the shape of the distribution.

Hide

```
hist(visits)
```



If we look at different samples of residents and their mean number of visits, how would the estimate change? What if our sample size was 100? 10,000? Let's investigate how the distribution of a collection of sample means changes as the sample size increases.

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```
mean.age <- function(n){
  trials = 1000
  my.samples <- matrix(sample(visits, size = n * trials, replace = TRUE), trials)

  means <- apply(my.samples, 1, mean)
  means
}
```

The function mean.age(n) performs 1000 times the experiment of drawing a sample of visits of size n at random with replacement from our dataset. This is called bootstrapping a sample from data.

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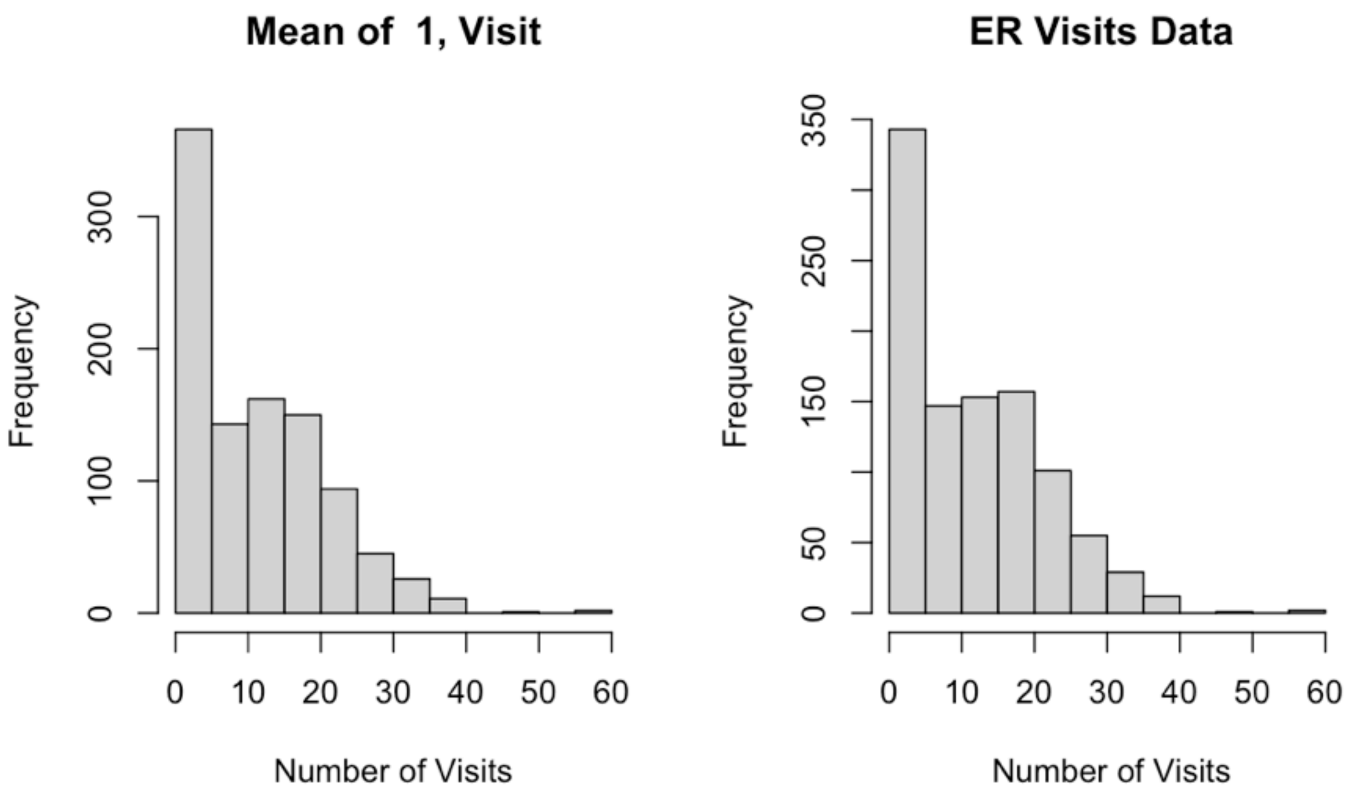
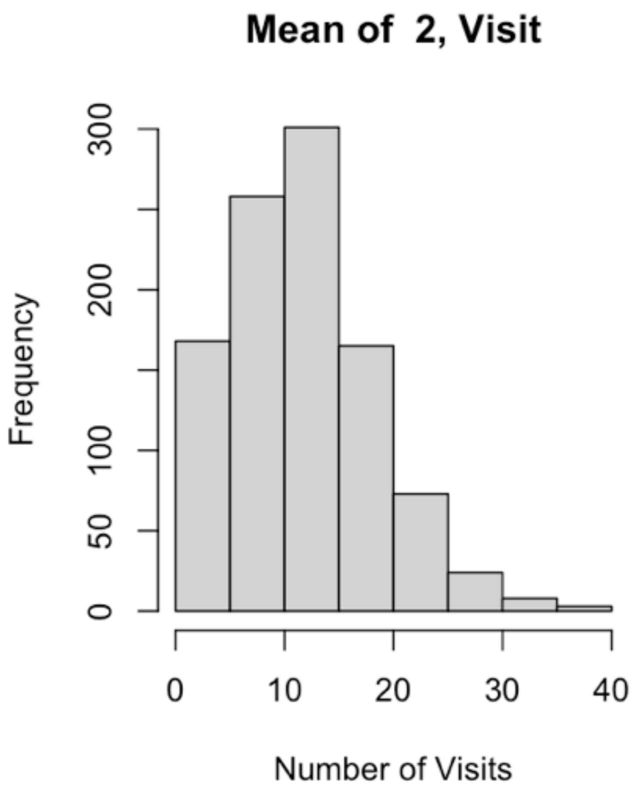
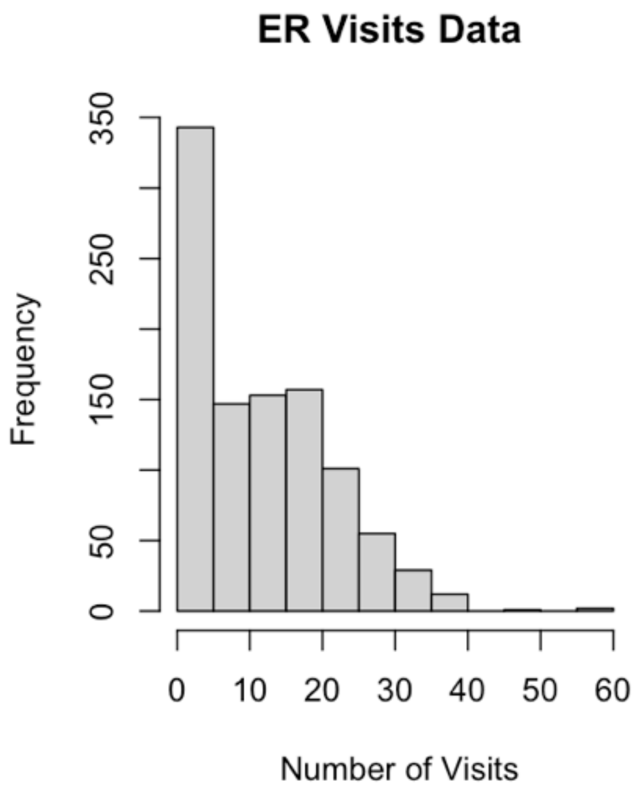
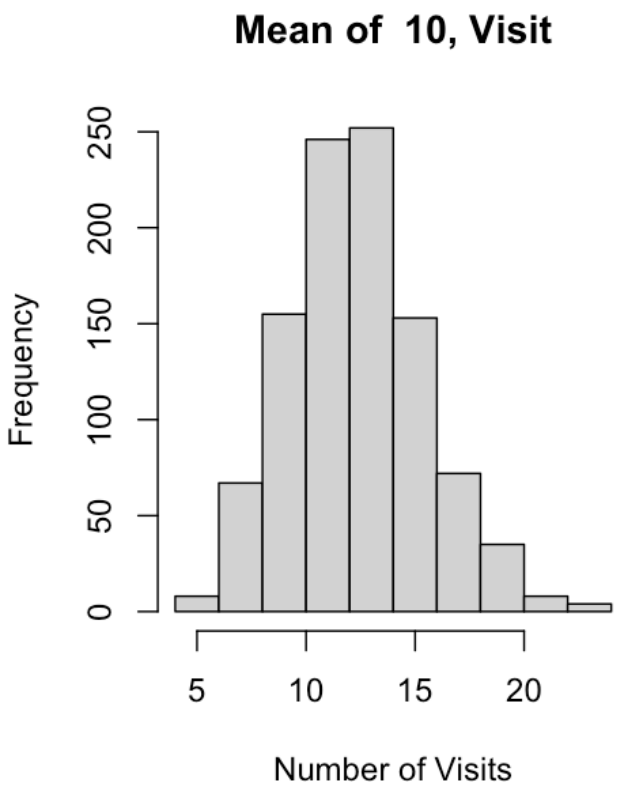
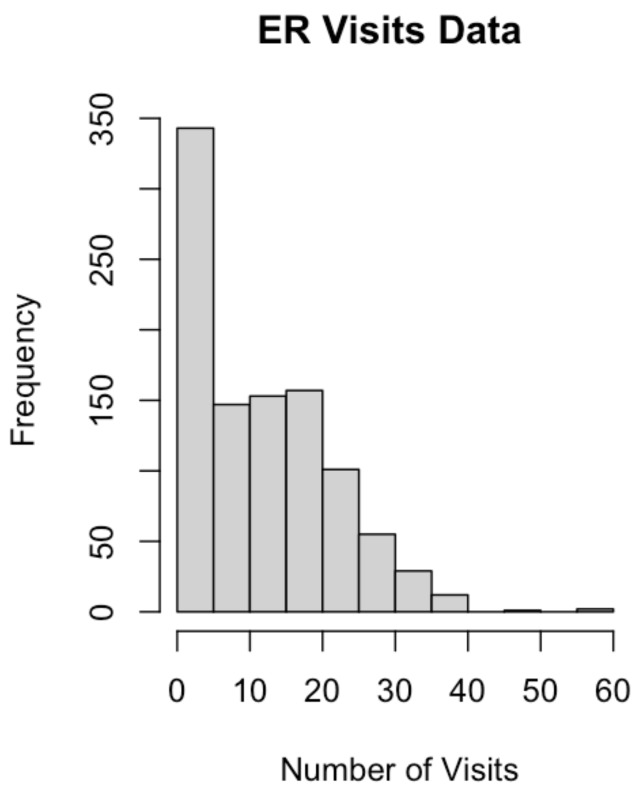
```
par(mfrow=c(1, 2))
means_variances = c(1, 2, 10, 20, 100, 200)

plots_mean_er =function(m) {
  hist(mean.age(m), main = sprintf("Mean of  %s, Visit", m), xlab="Number of Visit
s")
  hist(visits, main="ER Visits Data ", xlab=" Number of Visits")
}

mapply(plots_mean_er, means_variances)
```

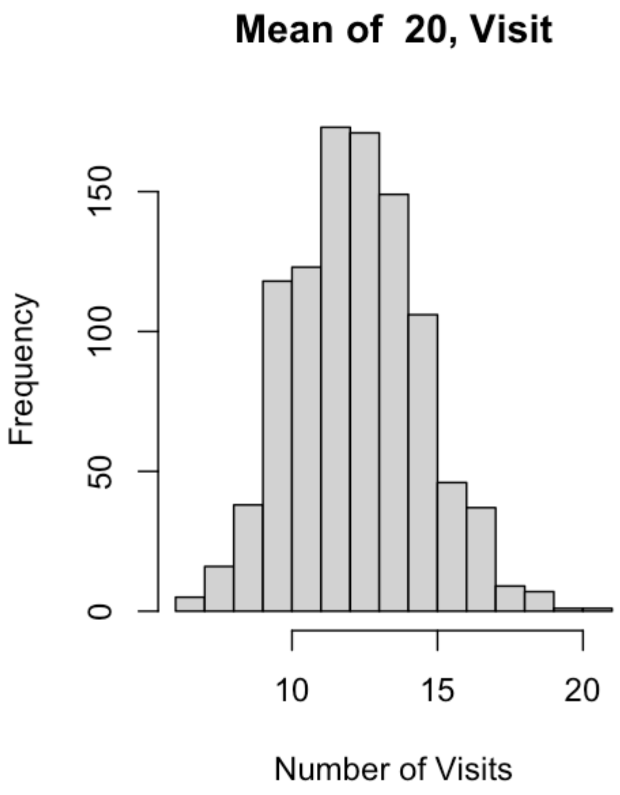


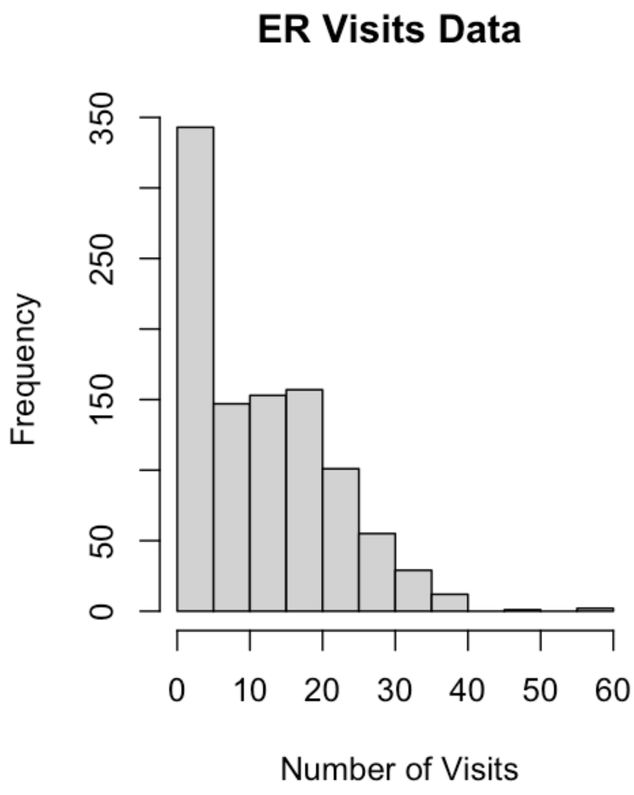

Mean of 2, Visit
Frequency
Number of Visits
0
50
100
200
300
0
10
20
30
40

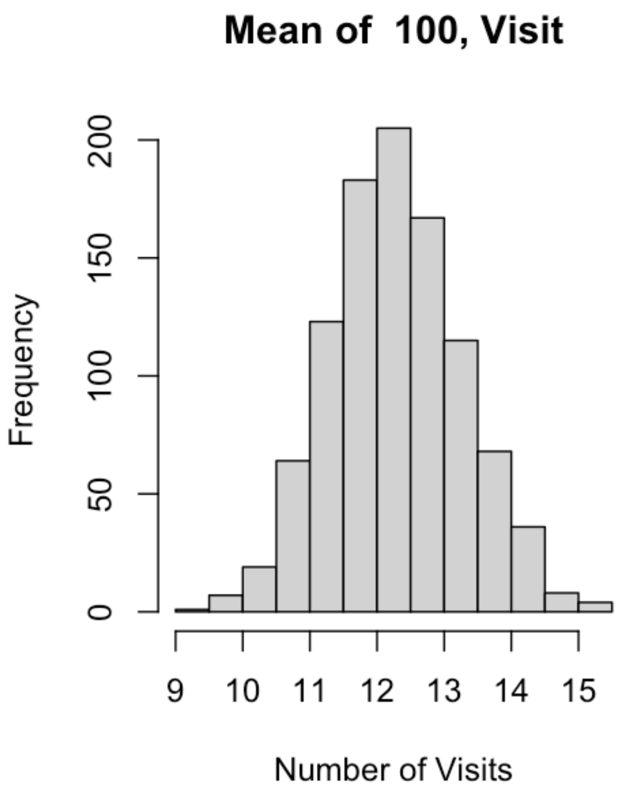

ER Visits Data
Frequency
Number of Visits
0
50
150
250
350
0
10
20
30
40
50
60

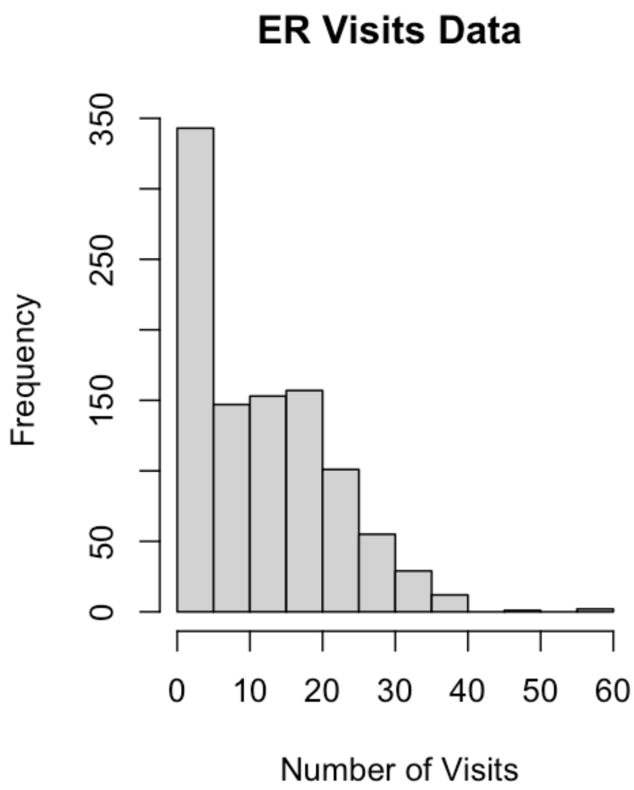

Mean of 10, Visit
Frequency
Number of Visits
0
50
100
150
200
250
5
10
15
20


ER Visits Data
Frequency
Number of Visits
0
50
150
250
350
0
10
20
30
40
50
60

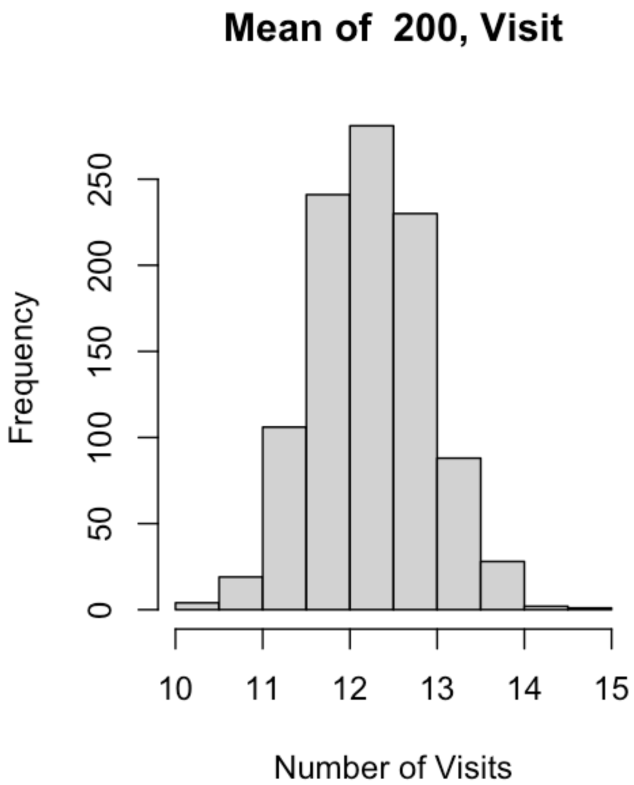

Mean of 20, Visit
Frequency
0
50
100
150
10
15
20
Number of Visits

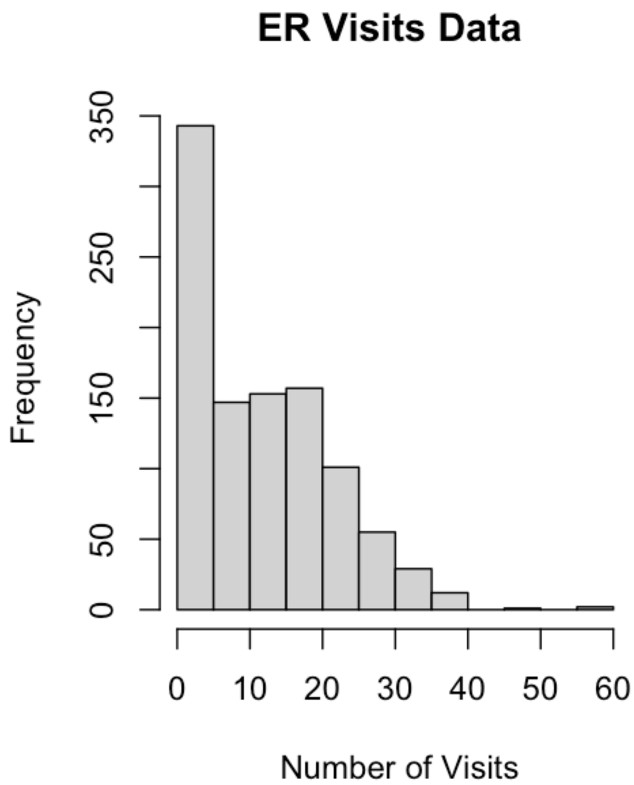

ER Visits Data
Frequency
0
50
150
250
350
0
10
20
30
40
50
60
Number of Visits


Mean of 100, Visit
Frequency
0
50
100
150
200
9
10
11
12
13
14
15
Number of Visits


ER Visits Data
Frequency
0
50
150
250
350
0
10
20
30
40
50
60
Number of Visits

```
          [,1]       [,2]       [,3]       [,4]       [,5]       [,6]
breaks    numeric,13 numeric,13 numeric,13 numeric,13 numeric,13 numeric,13
counts    integer,12 integer,12 integer,12 integer,12 integer,12 integer,12
density   numeric,12 numeric,12 numeric,12 numeric,12 numeric,12 numeric,12
mids      numeric,12 numeric,12 numeric,12 numeric,12 numeric,12 numeric,12
xname     "visits"   "visits"   "visits"   "visits"   "visits"   "visits"
equidist  TRUE       TRUE       TRUE       TRUE       TRUE       TRUE
```





VARIANCE OF X N

We see from the histograms that the distribution of the sample mean X n becomes narrower as n increases. In other words, the variance decreases as the sample size gets larger. Let's see if there's a functional relationship between the variance distribution of the sample mean X n and the sample size n. We can calculate the variances of mean.age(1), mean.age(2), mean.age(10), mean.age(20), mean.age(100), and mean.age(200) with a function called "vars."

Hide

```
variances <- data.frame(n=means_variances, variance = mapply(\(m) var(mean.age(m)),
means_variances))
variances
```

n	variance
<dbl>	<dbl>

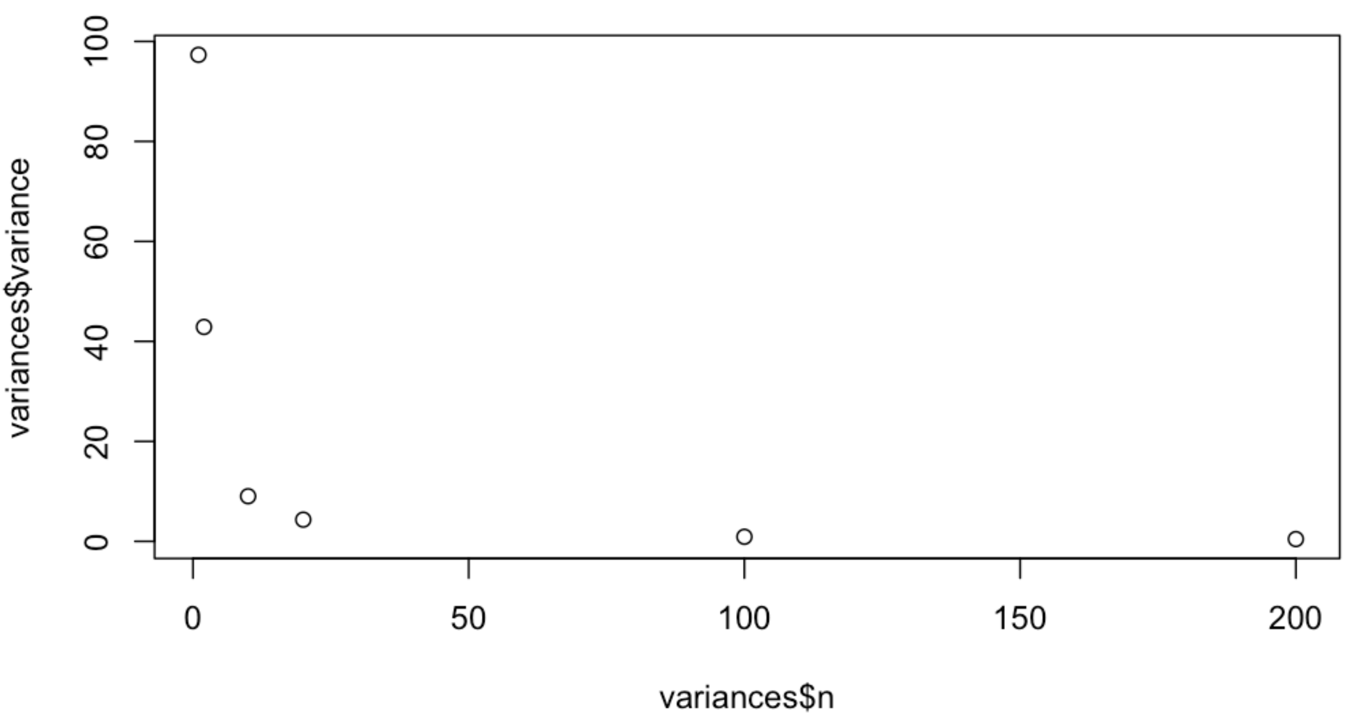
1	97.3230741
2	42.8905495
10	9.0283930
20	4.3397732
100	0.9192955
200	0.4543807

6 rows

Let's look at the relationship of variance with sample size n.

Hide

```
plot(variances$n, variances$variance)
```



In addition to becoming narrower, the shape of X n's distribution changes from being skewed to becoming symmetric and bell-shaped, like the normal distribution, as n becomes larger.

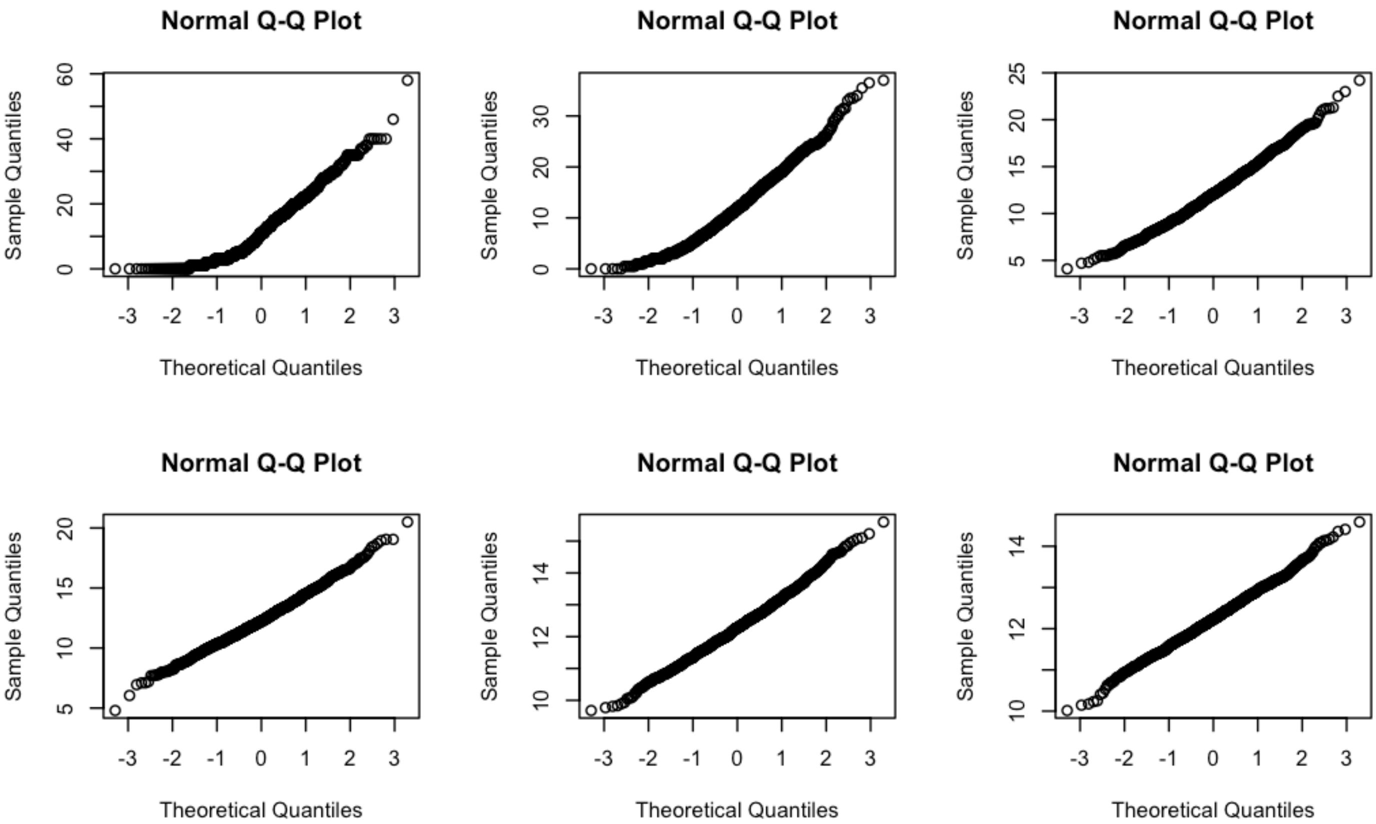
QUANTILE-QUANTILE PLOTS

We can test how close the distribution of X n is to the normal distribution by examining quantile-quantile (Q-Q) plots. In a Q-Q plot, quantiles of the sample are plotted against quantiles of a proposed distribution, also known as theoretical quantiles.

Hide

```
par(mfrow=c(2,3))
mapply(\(m) qqnorm(mean.age(m)), means_variances)
```

```
  [,1]         [,2]         [,3]         [,4]         [,5]         [,6]        
x numeric,1000 numeric,1000 numeric,1000 numeric,1000 numeric,1000 numeric,1000
y numeric,1000 numeric,1000 numeric,1000 numeric,1000 numeric,1000 numeric,1000
```



If the points of a Q-Q plot fall on a straight line, this indicates that the sample data are consistent with the proposed distribution (here, the normal distribution). As n increases, what changes in the Q-Q plots?

SHAPIRO-WILK TEST

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```
par(mfrow=c(2,3))
mapply(\(m) shapiro.test(mean.age(m)), means_variances)
```

```
          [,1]                          [,2]                         
statistic 0.923669                      0.9714733                    
p.value   4.237318e-22                  4.135191e-13                 
method    "Shapiro-Wilk normality test" "Shapiro-Wilk normality test"
data.name "mean.age(m)"                 "mean.age(m)"                
          [,3]                          [,4]                         
statistic 0.9917305                     0.9984986                    
p.value   2.155619e-05                  0.5537084                    
method    "Shapiro-Wilk normality test" "Shapiro-Wilk normality test"
data.name "mean.age(m)"                 "mean.age(m)"                
          [,5]                          [,6]                         
statistic 0.9986246                     0.9989437                    
p.value   0.6376499                     0.8427662                    
method    "Shapiro-Wilk normality test" "Shapiro-Wilk normality test"
data.name "mean.age(m)"                 "mean.age(m)"                
```

SAMPLING DISTRIBUTIONS

A statistic is a value computed from data. In our example, our statistic is the average number of ER visits.

Is a statistic a random variable? Recall that a random variable is a numeric value associated with the outcome of an experiment. It has a distribution, a mean (expected value), and a variance.

Because data constitute a random realization of an experiment, each time we run the experiment, we will get a different value for our statistic. So, in fact, it has a distribution, a mean (expected value), and a variance.

In other words, as our sample size increases, any distribution—normal or not—will have a mean that tends to behave normally.

THE CENTRAL LIMIT THEOREM

In the field of statistics, it's rarely possible to collect all of the data from an entire population. Instead, we can gather a subset of data from a population and then use statistics for that sample to draw conclusions about the population.

The 2 most common characteristics that we use to define a population are its mean and standard deviation. And when data comes from a normal distribution, we immediately have an idea of the data's shape.

The mean tells us the center of that distribution. The standard deviation tells us the spread.

The central limit theorem tells us that, no matter what the population distribution looks like, the distribution of the sample means will approach a normal distribution.